

KUSH VASHISTH

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EDUCATION

University of Southern California
Master's, Computer Science

Jan 2024 - Dec 2025
GPA: 3.81

Gujarat Technological University
Bachelor's, Computer Science

Jul 2019 - Jun 2023
GPA: 3.65

SKILLS

- **Skills:** Python, JavaScript, TypeScript, C/C++, SQL
- **Backend & Infra:** FastAPI, Django, Node.js, REST APIs, AsyncIO, Redis, AWS, Docker, CI/CD, Linux/Unix
- **Frontend:** React, Next.js
- **Data & ML:** Pytorch, Transformers, NumPy, Pandas, Elasticsearch
- **Systems / Patterns:** Distributed workers, Task queues, Background jobs, Agentic workflow design

PROFESSIONAL EXPERIENCE

University of Southern California
Research Assistant

Feb 2026 - Apr 2026
Los Angeles, CA, USA

- Implemented a natural-language banking assistant using LLMs through Ollama to process and interpret user banking queries.
- Built backend APIs with FastAPI and integrated a cloud database using Supabase to store user data, transaction records, and system outputs.
- Evaluated and refined LLM responses, database queries, and API workflows to improve accuracy of generated financial insights.

University of Southern California
Teaching Assistant | Advanced Mobile Devices & Games (Academic Role)

Jan 2025 - Aug 2025
Los Angeles, CA, USA

- Delivered C# and Unity lectures to 120+ students in a graduate-level game development course, supporting design and implementation
- Conducted weekly code reviews to improve gameplay mechanics, performance, and overall project structure
- Evaluated 15+ student projects across development phases from Alpha to Gold, providing technical feedback and scoring

MBA & Beyond
Software Engineer Intern

Feb 2022 - Jun 2022
Singapore

- Built a workflow automation app in SailsJS, reducing manual email response time by 30% and improving operational efficiency
- Refactored Squarespace site components and added analytics-based user tracking, improving overall engagement by ~20%
- Collaborated with design and content teams to enhance UI consistency across mobile and desktop experiences

PROJECTS & OUTSIDE EXPERIENCE

Agentic Workflow Orchestrator – Local AI Automation System

- Built a fullstack agentic workflow engine (FastAPI, React, Redis) with Python worker processes for async multi-step execution
- Integrated Gemma2-2B (via Ollama) converting natural language into validated JSON workflows with schema enforcement
- Implemented a distributed execution layer with parallel workers, dependency-aware chaining, and real-time task progress
- Added an extensible tool registry and HTML/text preprocessing pipeline to support robust and adaptable automation modules

MediHear – Real-Time Doctor-Patient Teleconsultation Platform

- Developed with Next.js, WebRTC, Socket.IO, and Firebase authentication to support secure remote consultations
- Implemented real-time speech transcription using the Web Speech API and a Flask NLP service for live symptom extraction
- Achieved ~300 ms end-to-end latency for video and transcription in local tests, enabling natural conversational flow

PII Detection and Masking

- Built an end-to-end PII detection pipeline using PyTorch and Transformers to identify 94 PII categories across large corpora
- Fine-tuned DeBERTaV3, achieving 99.56% F1 with a 27% reduction in false positives, improving precision in sensitive-data tasks
- Integrated Elasticsearch to support scalable indexing and sub-second querying of millions of anonymized documents

LEADERSHIP ROLES

CampTech

- Led a 9-person cross-functional team across instruction and operations to plan and deliver monthly programming workshops
- Organized and ran recurring free workshop series (8 sessions) covering web development, mobile apps, machine learning, and blockchain; supported ~80 total student attendees across sessions

Wellsprings Foundation(NGO)

- Founded and built the NGO's technical team, establishing ownership, roles, and execution cadence
- Led the NGO's digital expansion by delivering and maintaining core online assets and workflows to improve reach and operations